

VELIKO GRADISTE, Serbia

WMO: 132850

Lat: 44.750N

Lon: 21.517E

Elev: 276

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.1	7.4	8.1	17.2	12.4	10.4	20.5	26.1	29.2	23.1	32.4	4.1	110	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	93.9	69.6	90.6	69.2	87.5	68.4	73.5	85.0	71.9	84.0	70.5	82.5	5.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	112.5	77.8	68.4	105.5	76.1	66.8	99.6	74.7	37.2	84.8	35.7	83.9	34.6	82.5	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	18.9	16.6	DB	5.4	98.4	6.7	4.1	0.6	101.3	-3.3	103.7	-7.0	106.0	-11.9	109.0	
			WB	4.8	75.9	6.4	2.4	0.2	77.7	-3.5	79.1	-7.2	80.4	-11.8	82.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	33.0	36.5	44.6	54.7	62.8	69.2	72.8	72.7	63.9	54.2	45.0	35.4
	DBStd	15.84	8.04	8.71	8.13	7.18	6.43	6.23	5.58	5.75	6.84	7.81	8.29	7.66
	HDD50	1846	527	381	203	37	1	0	0	0	0	49	193	455
	HDD65	4859	992	797	631	314	120	29	6	8	101	342	601	918
	CDD50	3241	0	4	37	178	397	575	707	704	417	178	43	1
	CDD65	781	0	0	0	6	51	155	248	247	68	6	0	0
	CDH74	8337	0	0	2	88	518	1663	2720	2620	659	67	0	0
	CDH80	3306	0	0	0	12	101	611	1206	1175	195	6	0	0
Wind	WSAvg	5.6	6.3	6.6	6.6	6.5	5.1	4.2	4.0	4.3	4.9	6.1	7.3	6.0
Precipitation	PrecAvg	27.2	1.8	1.7	1.7	2.4	2.8	3.1	3.0	2.1	2.5	2.2	2.0	2.0
	PrecMax	36.3	3.4	3.5	4.7	4.4	5.6	7.0	7.8	6.1	6.5	5.6	6.0	5.0
	PrecMin	17.3	0.4	0.1	0.4	0.2	0.8	0.5	0.2	0.2	0.5	0.1	0.0	0.3
	PrecStd	5.4	0.9	1.0	1.0	1.1	1.5	1.7	1.9	1.5	1.4	1.4	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	63.6	72.2	81.5	86.5	94.1	98.9	98.4	91.3	80.3	69.0	56.1
		MCWB	46.9	50.2	54.6	60.1	65.2	69.6	70.9	71.0	66.9	62.4	55.6	48.2
	2%	DB	50.3	57.6	67.1	75.9	82.8	90.0	93.9	94.2	85.2	75.3	64.1	51.4
		MCWB	45.2	48.0	51.9	58.3	64.4	70.0	69.7	69.6	65.1	60.4	55.0	46.7
	5%	DB	46.9	52.7	62.5	71.7	80.0	86.9	90.5	90.5	81.2	71.2	60.2	48.4
		MCWB	43.4	45.9	50.2	56.2	63.6	68.9	69.4	68.9	63.8	58.7	53.0	44.8
	10%	DB	43.5	49.1	58.3	67.9	76.5	83.7	86.8	86.9	77.3	67.0	57.1	45.3
		MCWB	40.7	43.8	48.2	54.8	62.2	67.8	68.9	68.0	62.7	57.3	51.7	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	51.9	55.8	62.8	69.5	75.6	75.4	75.6	70.3	64.8	58.2	50.0
		MCDB	52.2	59.3	67.8	75.1	80.2	85.6	85.8	87.8	83.0	73.9	64.2	53.7
	2%	WB	46.0	49.1	53.2	60.0	66.9	72.9	73.2	73.2	67.6	62.1	56.1	47.3
		MCDB	49.6	55.7	63.8	71.9	77.6	83.8	85.4	84.5	79.0	71.4	61.9	50.9
	5%	WB	43.5	46.7	51.3	58.0	65.2	71.0	71.8	71.4	65.7	60.3	54.2	44.9
		MCDB	46.6	51.9	60.0	68.7	75.6	82.3	84.4	83.7	76.1	68.3	59.3	47.8
	10%	WB	41.0	44.3	49.2	56.1	63.7	69.3	70.4	69.9	64.2	58.5	51.9	42.7
		MCDB	43.3	48.9	57.1	65.6	73.5	80.2	82.6	82.1	74.0	65.8	56.7	45.2
Mean Daily Temperature Range	5% DB	MDBR	10.4	13.9	18.2	19.7	20.6	22.0	23.4	23.8	20.4	17.8	13.1	10.0
		MCDBR	16.0	22.7	26.9	27.9	26.5	27.6	29.8	30.0	27.2	24.8	18.5	14.5
		MCWBR	11.2	13.5	13.7	12.1	10.6	10.2	9.7	9.3	10.0	11.2	10.1	10.2
	5% WB	MCDBR	14.6	20.3	22.9	23.3	23.2	24.3	25.0	24.5	22.2	21.9	16.0	13.4
		MCWBR	10.6	12.5	12.2	10.9	10.6	10.6	10.2	9.7	9.3	11.0	9.4	9.9
Clear-Sky Solar Irradiance	taub	0.351	0.381	0.419	0.480	0.465	0.456	0.464	0.480	0.450	0.407	0.371	0.348	
	taud	2.276	2.175	2.145	2.000	2.104	2.174	2.155	2.104	2.150	2.259	2.308	2.310	
	Ebn at Noon	234	247	255	248	257	260	256	246	242	236	225	222	
	Edh at Noon	27	36	42	53	49	46	46	47	42	33	26	24	
All-Sky Solar Radiation	RadAvg	389	652	1039	1440	1739	1938	1957	1744	1219	826	501	328	
	RadStd	49	86	135	150	176	189	154	171	129	100	50	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.33	N/A	N/A	+1.78	+1.30	+1.60	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+1.19	N/A	N/A	N/A	+0.65	N/A	N/A	-270	+191	N/A

Nomenclature: See separate page